

agent-mem

Local memory service for coding agents

Source persistence, local search, and bounded session context

ORIGIN Research on long-term agent memory

SCOPE Capture, recall, and source references

TALK 8 minutes

The memory problem

DURING THE SESSION

Active context

During a session, the prompt, files, and tool results live in the active context.

AFTER THE SESSION

Project history is missing

The next session may not know earlier decisions. An old statement may be stale after it was corrected.

QUESTION How does a decision stay linked to its source?

Research origin

PAPER FINDINGS / ARCHITECTURE CONSEQUENCES

RAG / HippoRAG /
LongMemEval

Retrieval separates source, selection, and context handoff.

Generative Agents /
MemoryBank / MemGPT /
Reflexion

Original observations, derived experience, and session context are different data types.

A-MEM / Mem0

Updates need provenance and identity so older sources remain traceable.

These three findings define the current **agent-mem** core.

The agent-mem viewer

agent-mem

DASHBOARD

SOURCES

MEMORY RECORDS

SESSIONS / JOBS

GRAPH

AUDIT

View

All projects

Search sources...

LOCAL

SESSIONS

2

SOURCES

2

SPANS

1

RECORDS

0

JOBS

1

GRAPH SOURCES

2

TOKEN SAVINGS

97.3%

551 tokens saved

HEALTH

Local

read-only

RECENT SOURCES

2 loaded · click a row for raw evidence

TIME	HOST / SESSION	ROLE	EVIDENCE CLASS	STATE	SPANS	PREVIEW
2026-09-15T22:44:26.828Z	codex 79809d1c-c933-4...	USER	PROMPT	STORED	1	Pilot smoke test for the local Agent Mem viewer
2026-09-15T22:44:26.562Z	codex 79809d1c-c933-4...	SYSTEM	LIFECYCLE	STORED	0	-

SYSTEM STATUS

local runtime

capture

available

search

available

model

available

reranker

available

EVIDENCE REDUCTION

tokenizer

STORED BASELINE

566 tokens

SURFACED EVIDENCE

15 tokens

REDUCTION

97.3%

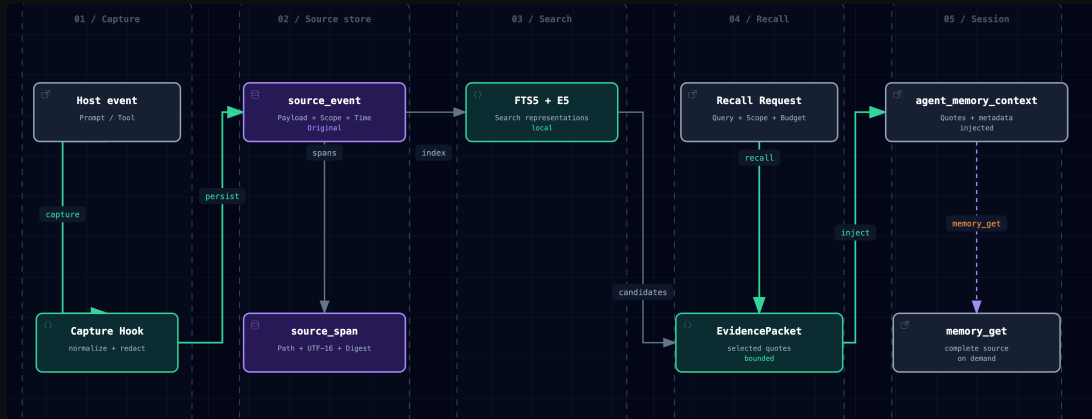
SAVED

551 tokens

agent-mem

4/ 8

Persistence: What gets stored

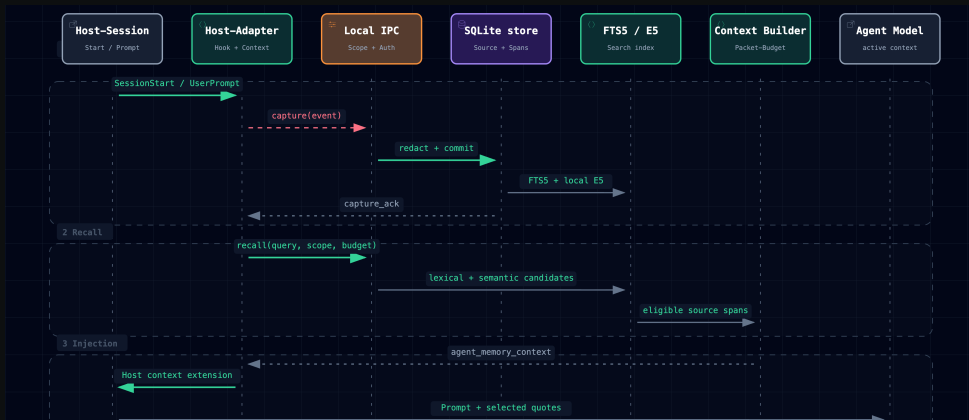


source_event: payload + scope + time
/ **E5:** local search representations

source_span: path + UTF-16 offsets + digest

FTS5

Session context: What enters the session



agent_memory_context: injection ID + source IDs + scope + status + selected quotes. The host receives a bounded packet.

Workflow in the Archify diagram

01 / CAPTURE

Host event

The capture hook cleans the event and writes `source_event` and `source_span`.

02 / RECALL

Candidates

FTS5 finds exact terms. E5 finds semantic variants. Scope and budget filter the selection.

03 / INJECTION

Session context

`agent_memory_context` enters the session.
`memory_get` loads the source when needed.

The diagram separates durable source data, candidate selection, and active session context.

Result and limits

CURRENT CORE

The store stays complete. The session stays bounded.

TODAY

`agent-mem` captures sources, searches locally, and returns only selected, source-grounded excerpts to the session.

NEXT EXTENSION

Revisions, temporal models, and evaluation for stale or conflicting evidence.

Every injected item points back to a stored source.